

1. Administrative & Study Identification

Full Study Title: A Phase 1b/2, Open-Label, Multicentre Study Assessing the Safety, Tolerability, Pharmacokinetics, and Preliminary Anti-tumor Activity of MEDI4736 in Combination With AZD9150 or AZD5069 in Patients With Advanced Solid Malignancies and Subsequently Comparing AZD9150 and AZD5069 Both as Monotherapy and in Combination With MEDI4736 as Second Line Treatment in Patients With Recurrent and/or Metastatic Squamous Cell Carcinoma of the Head and Neck **Short Title/Acronym:** SCORES Study (MEDI4736 + AZD9150/AZD5069 in Advanced Solid Tumors & RM-SCCHN) **Protocol Number:** D5660C00004 **NCT Number:** NCT02499328 **Sponsor Name:** AstraZeneca **Investigational Product:** MEDI4736 (durvalumab), AZD9150 (danvatirsen), AZD5069, and tremelimumab **Phase of Development:** Phase 1b/2 **Trial Status:** Completed **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** * *Part A (Dose-Escalation):* To assess the safety, tolerability, and define the Maximum Tolerated Dose (MTD) or Recommended Phase 2 Dose (RP2D) of MEDI4736 in combination with either AZD9150 or AZD5069. Additional cohorts evaluate combinations including tremelimumab. * *Part B (Dose-Expansion):* To evaluate the preliminary anti-tumor activity (Objective Response Rate [ORR]) of these combinations as a second-line treatment in recurrent/metastatic squamous cell carcinoma of the head and neck (RM-SCCHN). * **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Overall Survival (OS), Duration of Response (DoR), Pharmacokinetics (PK), and the immunomodulatory effects on the tumor microenvironment.

2.2 Background & Rationale To address the gap in treatment efficacy for patients with advanced solid malignancies and RM-SCCHN refractory to standard therapies. STAT3 (targeted by AZD9150) and CXCR2 (targeted by AZD5069) play major roles in immune suppression within the tumor microenvironment. Combining these targeted agents with the anti-PD-L1 immune checkpoint inhibitor durvalumab (MEDI4736), and potentially tremelimumab, is hypothesized to stimulate T-cells, overcome resistance to PD-L1 monotherapy, and improve clinical response rates.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (Comparing monotherapy to combination therapies)

in Part B) * **Blinding:** Open-label * **Randomization:** Randomized (1:1 allocation to different treatment arms during the Part B dose-expansion phase)

3.2 Planned Enrollment & Duration * **Target N:** 340 evaluable patients * **Number of Sites:** Multicenter (Global) * **Estimated Study Start:** 06-Aug-2015 * **Estimated Primary Completion:** 28-Feb-2020 (Overall Study Completion: 08-Oct-2025)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female patients ≥ 18 years of age. 2. Eastern Cooperative Oncology Group (ECOG) Performance Status (PS) score of 0 or 1. 3. Measurable disease, defined as at least 1 lesion ≥ 10 mm by CT scan per RECIST v1.1 (lymph nodes ≥ 15 mm short axis). 4. *Part A Specific:* Histological confirmation of a solid malignancy (other than HCC) refractory to standard therapy or lacking standard of care, including specific cohorts for castrate-resistant prostate cancer. 5. *Part B Specific:* Histologically confirmed RM-SCCHN not amenable to curative therapy, having failed at least 1 prior platinum-based chemotherapy (depending on cohort).

4.2 Exclusion Criteria 1. Clinically active brain metastases or spinal cord compression. 2. Prior exposure to AZD9150, AZD5069, MEDI4736, or any other anti-PD-1/PD-L1 antibody (except for specific anti-PD-L1 pre-treated cohorts in Part B). 3. Active or prior documented autoimmune disease, or use of immunosuppressive medication within 14 days before the first dose of study drugs. 4. Second malignancy other than SCCHN, or history of treatment for invasive cancer in the past 3 years.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Durvalumab (MEDI4736) administered at 1500 mg (or weight-based equivalent) once every 4 weeks, in combination with either AZD9150 (2-3 mg/kg weekly) or AZD5069. Tremelimumab is also administered in fixed-dose combination with MEDI4736 and AZD9150/AZD5069 in specific Part A cohorts. * **Route of Administration:** Intravenous (IV) for durvalumab, AZD9150, and tremelimumab; Oral (PO) for AZD5069. * **Duration of Treatment:** Until documented disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard supportive care medications, palliative local radiotherapy (under specific protocol conditions). * **Prohibited:** Any concurrent investigational agents, live attenuated vaccines within 30 days of first dose, immunosuppressive doses of systemic corticosteroids, and other systemic anti-cancer therapies.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Baseline Tumor Imaging (CT/MRI), Baseline Labs |
| **Baseline** | Day 1 | Randomization (Part B), First Dose of Investigational Product, PK/Biomarker Sampling | | **Interim** |
Every 2 to 4 Weeks | Safety Labs, PK/PD Sampling, IP Administration. Tumor assessments via CT/MRI every 8 weeks (Q8W)
| | **End of Study** | Disease Progression | End of Treatment Visit, Safety Follow-Up (30 Days post-last dose), Survival Follow-Up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** *Part A:* Incidence of Dose-Limiting Toxicities (DLTs) and Adverse Events (AEs) to define the MTD using a Continual Reassessment Method (CRM). *Part B:* Objective Response Rate (ORR) defined as the percentage of patients with a confirmed Complete Response (CR) or Partial Response (PR). * **Measurement Method:** Clinical safety evaluation per CTCAE criteria and Investigator-assessed tumor imaging per RECIST v1.1.

7.2 Secondary & Exploratory Endpoints * **Pharmacokinetics (PK):** Specifically, AZD9150 AUC_{0-6h} and C_{max} at Lead in Day -7; AZD9150 AUC_{0-6h} at Cycle 2 Day 1. * Progression-Free Survival (PFS) and Overall Survival (OS). * Duration of Response (DoR) and Disease Control Rate (DCR) at 12 weeks. * Evaluation of immunomodulatory biomarkers in the tumor microenvironment (e.g., PD-L1 expression, immune cell infiltrate).

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous monitoring at every study visit, graded utilizing the NCI CTCAE v4.03/v5.0.
 - **Serious Adverse Events (SAEs):** Must be reported to the Sponsor within 24 hours of site awareness.
 - **Data Monitoring Committee (DMC):** A Safety Review Committee (SRC) evaluates dose-escalation cohorts to determine the MTD/RP2D prior to initiating dose expansion.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for efficacy outcomes; Safety Analysis Set for all patients receiving at least one dose of study medication.
- **Sample Size Justification:** Target of 340 patients, powered to detect meaningful improvements in ORR between combination and monotherapy cohorts.

- **Handling of Missing Data:** Standard oncology data imputation methods; patients without post-baseline evaluations are considered non-responders for ORR.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP), applicable regulatory requirements, and the Declaration of Helsinki.
 - **Monitoring Plan:** Standard risk-based onsite and remote monitoring of source data, focusing on safety and RECIST assessments.
 - **Audit Trail:** Validated eCRF systems with automated audit trails and data clarification form mechanisms.
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11. Study Identifiers & Governance

Primary ID: D5660C00004 **Registry Number:** NCT02499328 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** SCORES Study (MEDI4736 + AZD9150/AZD5069) **Official Regulatory Title:** A Phase 1b/2, Open-Label, Multicentre Study Assessing the Safety, Tolerability, Pharmacokinetics, and Preliminary Anti-tumor Activity of MEDI4736 in Combination With AZD9150 or AZD5069 in Patients With Advanced Solid Malignancies and Subsequently Comparing AZD9150 and AZD5069 Both as Monotherapy and in Combination With MEDI4736 as Second Line Treatment in Patients With Recurrent and/or Metastatic Squamous Cell Carcinoma of the Head and Neck

12. Clinical Framework (Oncology/SCCHN Specific)

Primary Condition: Advanced Solid Tumors & Metastatic Squamous Cell Carcinoma of the Head and Neck **Diagnosis Threshold:** Refractory to standard therapy; radiologically measurable disease ≥ 10 mm per RECIST v1.1 **Medical Methodology:** Combined Immune Checkpoint Blockade (anti-PD-L1) and Microenvironment Modulation (STAT3 / CXCR2 inhibition) **Optimization Target:** Tumor shrinkage and prolongation of overall survival through T-cell stimulation

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for advanced solid malignancies and RM-SCCHN **Education Protocol (HCP):** Site Initiation Visits (SIV) and comprehensive training on investigational product administration, safety management, and RECIST evaluation **Education Protocol (Patient):** Informed consent detailing potential immunotherapy-related adverse events and compliance expectations **Quality Audit Mechanism:** Centralized

imaging review (as applicable), source data verification (SDV), and continuous pharmacovigilance tracking

14. Observation & Follow-Up Schedule

Total Duration: Treatment until disease progression; followed by extended overall survival follow-up **Onsite Visit Cadence:** Every 2 to 4 weeks during the active treatment phase **Remote Monitoring Frequency:** Survival follow-up contacts every 2 months after treatment discontinuation **Checklist Review Interval:** Tumor imaging assessments (CT/MRI) conducted every 8 weeks (Q8W)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				